Worksheet: the unbalanced solver and the glymphatic DCE-MRI problem

Companion to `python/optimal_transport/*_unbalanced.py`, `docs/unbalanced_1d.md`, and Chen, Benveniste, Tannenbaum, Sci. Rep. 14 (2024)

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Goal for today. Be able to (i) write the unbalanced problem and say what δ does, (ii) derive both LADMM subproblems, the prox and the projection, and see that each is a one-line change to the balanced code, (iii) say what every symbol means in a rat-brain tracer scan and translate urOMT's parameters into ours, (iv) build ρ_0, ρ_1 from the public data, and (v) know the three edits that port the 1D solver to 3D. Parts A–C are pen, paper, and the 1D numpy code (runs on the Mac). Parts D–F are the physics, the data recipe, and the GPU plan. Answers in Appendix A.

Conventions

Balanced solver: $\min \iint |m|^2/2\rho$ s.t. $\partial_t \rho + \nabla \cdot m = \varepsilon \Delta \rho$, $\rho(0) = \rho_0$, $\rho(1) = \rho_1$, $m \cdot n = 0$ on the walls of $[0,1]^d$, $t \in [0,1]$. LADMM splitting (`pipeline.py`): $x = (\rho, m)$ on the staggered grid, $y = (\rho, m)$ on cell centres, constraint $Ax - y = 0$ with A the staggered-to-centre interpolation; $f_1(x) = \iota_{\mathrm{FP}}(x)$ handled by a projection, $f_2(y) = \mathrm{KE}$ handled by a pointwise prox. Chen et al. write $\rho|v|^2$ with no $\frac{1}{2}$ and $\alpha\chi\rho r^2$ for the source; we keep our $\frac{1}{2}$ throughout and translate in Part D.

Part A. The unbalanced formulation

Exercise 1 *Write the problem.*

Add a source $\zeta(t,x)$ to the continuity equation and charge for it with the Fisher–Rao cost. Write out

$$\min_{\rho,m,\zeta} \iint \frac{|m|^2}{2\rho} + \frac{\delta^2}{2}\frac{\zeta^2}{\rho}\,dx\,dt \quad \text{s.t.} \quad \partial_t\rho + \nabla\cdot m - \varepsilon\Delta\rho = \zeta, \;\; \rho(0)=\rho_0,\; \rho(1)=\rho_1,$$

with $\int \rho_0 \neq \int \rho_1$ allowed. Introduce the growth rate $g = \zeta/\rho$ and the velocity $v = m/\rho$ and rewrite the integrand as $\frac{1}{2}\rho\,(|v|^2 + \delta^2 g^2)$. What are the units of δ? Why must δ have those units for the two terms to be comparable?

Exercise 2 *Pure reaction has a closed form.*

Suppose $\rho_1 = c\,\rho_0$ for a constant $c > 0$ and $\varepsilon = 0$. Guess that the optimum has $m \equiv 0$ and find the path. With $m = 0$ the constraint is $\partial_t\rho = \zeta$, so the cost is $\frac{\delta^2}{2}\iint (\partial_t\rho)^2/\rho$. Substitute

$\rho = \sigma^2$ and minimize $2\delta^2 \iint (\partial_t \sigma)^2$ pointwise in x: show σ is linear in t, so

$$\rho_t = \big((1-t)\sqrt{\rho_0} + t\sqrt{\rho_1}\big)^2, \qquad \text{cost} = 2\delta^2 \int \big(\sqrt{\rho_1} - \sqrt{\rho_0}\big)^2 dx = 2\delta^2(\sqrt{c}-1)^2 \int \rho_0.$$

This is the Hellinger geodesic. It is test T1 of `unbalanced_1d.md`: for uniform $\to 2\times$uniform on $[0,1]$ the cost is $2\delta^2(\sqrt{2}-1)^2$, which with $\delta = 0.3$ is 0.03088; the solver hit it to all printed digits in 53 iterations. Why does adding $\varepsilon > 0$ not change this answer when ρ_0 is uniform?

Exercise 3 *What δ decides: move or teleport.*

Two unit-mass lumps a distance d apart, $\varepsilon = 0$. Compare the cost of (a) transporting the mass, $\frac{1}{2}d^2$, with (b) destroying it in place and recreating it at the target, which by Exercise 2 costs $2\delta^2(0-1)^2 + 2\delta^2(1-0)^2 = 4\delta^2$. Show the crossover is at $d \propto \delta$ and that the true WFR geodesic (which shrinks the mass while moving it) is cheaper than both, so the exact crossover is a pure number times δ; the hard limit beyond which no transporting path competes is $\pi\delta$. The 1D tests anchor this: T3 with gap $0.19 > \pi\delta = 0.157$ teleports (kinetic energy $5 \cdot 10^{-6}$), T3′ with $\pi\delta = 0.31 > 0.19$ transports (kinetic energy 0.0126, mass dips to 0.56 mid-way). Explain the mass dip in T3′ in words.

Exercise 4 *Optimality conditions.*

Introduce a multiplier $\phi(t,x)$ for the constraint and derive, pointwise where $\rho > 0$,

$$v = \nabla\phi, \qquad \delta^2 g = \phi \quad \Longrightarrow \quad m = \rho\nabla\phi, \;\; \zeta = \rho\phi/\delta^2.$$

Conclude the consistency relation $v = \nabla(\delta^2\zeta/\rho)$ between the velocity and the growth rate, which is test T5 (median relative error $8 \cdot 10^{-5}$). What happens to this relation as $\delta \to \infty$, and why does that recover the balanced problem (test T4: difference to the balanced solver $\propto 1/\delta^2$)?

Part B. The two subproblems

The code's trick is $s := \delta\zeta$. Then the reaction cost is $s^2/2\rho$ and the constraint is $\partial_t\rho + \nabla\cdot m - \varepsilon\Delta\rho - s/\delta = 0$.

Exercise 5 *The prox is the old prox.*

The y-subproblem is, pointwise at each cell,

$$\min_{\rho,m,s} \; \frac{|m|^2 + s^2}{2\rho} + \frac{1}{2\sigma}\big(|\rho-\bar\rho|^2 + |m-\bar m|^2 + |s-\bar s|^2\big), \qquad \sigma = 1/\gamma.$$

(a) Differentiate in m and in s and show both shrink by the same factor, $m = \frac{\rho}{\rho+\sigma}\bar m$, $s = \frac{\rho}{\rho+\sigma}\bar s$.
(b) Substitute back, differentiate in ρ, and show ρ solves the cubic

$$\rho^3 + (2\sigma - \bar\rho)\rho^2 + (\sigma^2 - 2\sigma\bar\rho)\rho - \sigma\big(\sigma\bar\rho + \tfrac{1}{2}q^2\big) = 0, \qquad q^2 = |\bar m|^2 + \bar s^2.$$

(c) Compare with the balanced cubic in `prox.py`: the only difference is q^2. Why is the function $(\rho, m, s) \mapsto (|m|^2 + s^2)/2\rho$ convex, and what is it the perspective of? (d) Which cost would *not* reduce to this cubic? Give one example (an entropy-type or L^1 growth penalty) and say what the pointwise solve would become.

Code check: `prox_unbalanced.py`

`q2 = mx_c**2 + s_c**2`; coefficients `b = 2*sigma - rho_c`, `c = sigma**2 - 2*sigma*rho_c`, `d = -sigma*(sigma*rho_c + 0.5*q2)`; `fac = rho_new/(rho_new + sigma)`; `s_new = fac*s_c`. Match (a), (b). The balanced `solve_cubic` is reused unchanged. In 3D, `q2` gains `my**2 + mz**2`.

Exercise 6 *The projection is the old projection plus a constant.*

The x-subproblem is the Euclidean projection of $(\bar{\rho}, \bar{m}, \bar{s})$ onto the affine set $\{Mx = 0\}$, $M = [\, A_{\mathrm{FP}},\ -I/\delta \,]$, where $A_{\mathrm{FP}}(\rho, m)$ is the discrete $\partial_t \rho + \nabla \cdot m - \varepsilon \Delta \rho$ (with ρ_0, ρ_1 baked into the time differences) and the $-I/\delta$ block acts on s. (a) Write the projection as $x = \bar{x} - M^\top \varphi$ with $MM^\top \varphi = M\bar{x}$, and show $MM^\top = A_{\mathrm{FP}} A_{\mathrm{FP}}^\top + \delta^{-2} I$. (b) Hence the correction is: (ρ, m) get $A_{\mathrm{FP}}^\top \varphi$ exactly as in the balanced code, and s gets $s = \bar{s} + \varphi/\delta$. Check the sign. (c) The balanced code solves $A_{\mathrm{FP}} A_{\mathrm{FP}}^\top \varphi = f$ by a DCT in space, which decouples the spatial modes k, followed by one tridiagonal (CN) or ETD system in time per mode. Explain why adding $\delta^{-2} I$ adds δ^{-2} to the diagonal of every one of those time systems and changes nothing else. (d) The DC mode. For $k = 0$ (spatially constant mode) the balanced operator is the Neumann Laplacian in time, which is singular: its null vector is the constant in t, and solvability requires the residual's DC mode to sum to zero over t, which is mass conservation. Show that $+\delta^{-2}$ makes it regular, and interpret: the total mass at each time slice is now a free variable, paid for through s.

Code check: `projection_unbalanced.py`

`inv_d2 = 1/delta**2` added to `D` (interior rows) and to `D[0]`, `D[-1]`; `dc_den = problem.lambda_t[:,0] + inv_d2`; the residual `f` has `- s/delta`; the DC mode is solved by a DCT in t divided by `dc_den`; `s_out = s + phi/delta`. Match (a)–(d). Confirm the per-mode off-diagonal `e` is untouched.

Exercise 7 *Why the LADMM loop itself does not change.*

The coupling operator A is the interpolation on (ρ, m) and the *identity* on s (`A_fn`, `At_fn` in `pipeline_unbalanced.py`). Why does this keep $\|A\| \le 1$, and why does that matter for the step sizes γ, τ and the stopping rule? Where does s live on the grid, and why there? (Hint: where does the FP residual live?)

Exercise 8 *Cost per iteration.*

Count arrays and kernels. In d dimensions the balanced state has $1 + d$ fields; the unbalanced one has $2 + d$. The projection does the same DCT and the same batched time solve, plus one cheap DC-mode solve. Estimate the relative increase in memory and in elementwise work in 3D. Why can the iteration count go *down* (T1: 53 iterations), and in which regime does it go up (T3)?

Part C. Code audit checklist (1D)

1. `StateU`: third field `s` of shape `(nt, nx)` in both x and y; `norm` includes it. `pipeline_unbalanced.norm_fn` weights it by $dt\,dx$ like ρ, m: is s a density-scale quantity? ($s = \delta\zeta$ has units of ρ times length-/time; on the unit box with $\delta \sim 1$ it is comparable to m.)

2. Initial guess: ρ linear in t between ρ_0, ρ_1, $m = s = 0$. With unequal masses the linear interpolant violates the balanced constraint by a constant-in-x amount; which variable absorbs it on the first projection?

3. Dual `solve_y`: `prox_ke_reaction_cc(z_hat - delta_mult/gamma, 1/gamma)`. The name `delta_mult` is the LADMM multiplier, not the WFR δ. Rename mentally.

4. Diagnostics returned: `ke`, `reaction`, `mass_t`. Test T6: the discrete mass balance $\frac{d}{dt}\int \rho = \int s/\delta$ holds to 10^{-13}. Reproduce it from `mass_t` and `s_stag`.

5. Run `scripts/validate_unbalanced.py` once and read every printed line against the table in `unbalanced_1d.md`.

Part D. What the symbols mean in a rat-brain tracer scan

The experiment (Chen et al. 2024, Methods; details not stated there are flagged)

A 3-month-old healthy rat, anaesthetized. The MRI contrast agent gadoteric acid (Gd-DOTA) is infused into the cerebrospinal fluid, in this lab's protocol via a cannula in the cisterna magna at the base of the skull. T1-weighted 3D images are acquired every 5 minutes for 140 minutes, 29 frames. Gadolinium shortens T_1, so the signal rises where tracer is; the authors convert each frame to *percent signal change from the pre-injection baseline* and treat that as proportional to tracer concentration. The analysis uses every other frame, 15 inputs 10 minutes apart (repository file names `images_7_tind_1` to `images_35_tind_15`), restricted to a brain mask, on a $56 \times 106 \times 51$ grid, smoothed (`smooth=1` in the run folder). Field strength, voxel size and the mask construction are not stated in the paper; read them off the data files. Successive pairs are solved in a chain, the interpolated end state of one pair seeding the next.

Exercise 9 *Dictionary.*

Fill in what each of our symbols is in the scan, then compare with the answers.

ours	in the scan	caution
$\rho(t, x)$	_____	linear in Gd only at low concentration
$\int \rho\, dx$	_____	arbitrary units; only the *ratio* M_1/M_0 is physical
$v = m/\rho$	_____	
ε	_____	see Exercise 11
$\zeta = \rho g$, $g > 0$	_____	
$g < 0$	_____	
δ	_____	
$t \in [0, 1]$	_____	

Exercise 10 *urOMT to ours: the source.*

Their model: $\partial_t \rho + \nabla \cdot (\rho v) = \sigma \Delta \rho + \chi \rho r$ with cost $\iint \rho |v|^2 + \alpha \chi \rho r^2$, no $\frac{1}{2}$, $\chi \in \{0, 1\}$ an indicator of where sources are allowed, $\chi \equiv 1$ for the rat. Match $\zeta = \chi \rho r$, $g = r$ where $\chi = 1$, and show that dividing their integrand by 2 gives ours with

$$\delta^2 = \alpha \qquad \text{(in their units, voxels}^2\text{)}.$$

Now nondimensionalize. Their grid has $\Delta x = 1$ voxel and a pair of frames spans $T = m\,\Delta t = 10 \times 0.4 = 4$ time units. Rescaling $x \to x/L$, $t \to t/T$ with L the box side in voxels, show $v \to vT/L$, $g \to gT$, and hence

$$\delta_{\text{ours}} = \sqrt{\alpha}/L.$$

Evaluate for the rat ($\alpha = 10^4$, $L = 106$, the longest axis) and for the spheres ($\alpha = 9000$, $L = 50$). Which regime of Exercise 3 is this? What does that say about how the optimizer will use the source: to teleport mass across the brain, or to adjust mass locally where ρ is already large?

Exercise 11 *urOMT to ours: the diffusion, and is it physical?*

Their $\sigma = 0.002$ voxel^2 per time unit. Show $\varepsilon_{\text{ours}} = \sigma T/L^2$ and evaluate for $L = 106$ and $L = 50$. Then put units back. Assume a voxel of 0.3 mm (typical for this lab's 9.4 T rat protocol; confirm from the data) and one time unit $= 10\,\text{min}/4 = 150$ s. Convert σ to mm^2/s. Compare with the free diffusivity of Gd-DOTA in water at body temperature, about $4 \cdot 10^{-4}$ mm^2/s, reduced by tissue tortuosity to roughly $1.5 \cdot 10^{-4}$ mm^2/s. By what factor is their σ below the physical value? Finally compute the *physical* ε for one 10-minute pair on the $L = 106$ crop. Is it inside the range the 3D suite has already run (10^{-3}) and the 2D suite (10^{-3} to 10^{-1})? Which of the two values would you run, and why both?

Exercise 12 *Where mass comes and goes.*

Over a 10-minute pair the total tracer in the field of view changes because (a) tracer is still arriving from the infusion site, (b) tracer leaves along perineural and meningeal lymphatic routes out of the field of view and is cleared to blood, (c) the signal-to- concentration map saturates at high concentration near the injection site, so early frames *under*count mass there, (d) noise and partial volume. Which of these is a genuine source ζ in the sense of the model, and which is a measurement artefact the model will nevertheless absorb into ζ? Given the Fisher–Rao cost ζ^2/ρ, where will the optimizer prefer to place mass gain and loss, and does that match the paper's observation of early $r > 0$ in CSF spaces and later $r < 0$ along the skull base?

Exercise 13 *Hard endpoint versus their free endpoint.*

They solve with $\rho(0) = \rho_0^{\text{img}}$ fixed and a penalty $\beta\|\rho(T) - \rho_1^{\text{img}}\|^2$, with $\beta = 50$ for the rat (versus 5000 for the clean spheres). We impose $\rho(1) = \rho_1$ exactly. (a) With a small β their end state may differ visibly from the image; with ours the image's noise is imposed exactly and must be produced by v and ζ. Which quantity absorbs pixel noise in our formulation, and at what cost scale (use ζ^2/ρ with $\zeta \sim$ noise, $\rho \sim$ local level)? (b) Argue that smoothing ρ_0, ρ_1 with the same kernel is the cheap substitute for a soft endpoint. (c) What would a soft endpoint change in our LADMM: the prox, the projection, or the coupling? (Only the terminal row of A_{FP}: ρ_1 becomes an unknown with a quadratic penalty, so the time systems get a modified last row. It is not part of today.)

Part E. Setting up ρ_0 and ρ_1

1. **Pick the pair.** Two consecutive inputs, 10 minutes apart. Start where the mass change is clearly nonzero and the geometry is simple: an early pair (tracer entering, $M_1 > M_0$) and a late pair (clearance, $M_1 < M_0$). Record M_0, M_1 in the data's units before doing anything else.

2. **Values.** Percent signal change, masked to the brain. Clip negatives to zero (noise below baseline), then add a floor of 10^{-3} of the peak inside the mask and a smaller floor outside it, so the density is strictly positive everywhere on the box. The prox tolerates $\rho \to 0$, but the hard endpoint with $\rho_1 = 0$ where $\rho_0 > 0$ forces that mass to be destroyed, which is a real statement; make it only inside the mask.

3. **Smoothing.** One Gaussian pass, same kernel on both frames, width about one voxel; this is what their `smooth=1` does. It is also the substitute for their soft endpoint (Exercise 15).

4. **Geometry.** The crop is $56 \times 106 \times 51$ voxels, isotropic. Our 3D setup takes separate n_x, n_y, n_z but one side length L, so unequal counts would stretch the voxels. Pad the volume with the outside floor to a cube of 106^3 voxels, or downsample by two to $28 \times 53 \times 26$ and pad to 56^3 (or 64^3) for the first run. Padding with near-zero density costs nothing in transport and keeps the Neumann walls away from the tracer. Their boundary is also a Neumann box, not the brain surface.

5. **Scale.** Divide both frames by M_0 so that $\sum \rho_0 \, dV = 1$ and $\sum \rho_1 \, dV = M_1/M_0$. The mass ratio is the physical content; the overall scale sets the density peak, which sets γ (the balanced 2D/3D runs want $\gamma \approx 0.1$ at peaks of order 10–60). Do *not* renormalize ρ_1 to unit mass: that is exactly what the balanced code does and what the unbalanced model exists to avoid.

6. **Parameters.** $\delta = \sqrt{\alpha}/L$ and $\varepsilon = \sigma T/L^2$ from Exercises 12 and 13 with L the padded cube side in voxels, so the first run reproduces their setting; then $\varepsilon_{\text{phys}}$ as a second run.

7. **Conversion back.** Velocity in voxels/min $= v\,L/T_{\min}$ with $T_{\min} = 10$; multiply by the voxel size for mm/min. Growth rate in 1/min $= g/T_{\min}$; their r-map is our $s/(\delta\rho)$ averaged over t.

Exercise 14 *Sanity numbers before the run.*

From ρ_0, ρ_1 alone, compute: M_1/M_0; the Hellinger cost $2\delta^2 \int (\sqrt{\rho_1} - \sqrt{\rho_0})^2$ (the cost of pure reaction, an upper bound on the optimum); the balanced W_2 cost of the renormalized pair from a cheap Sinkhorn or from the solver's balanced mode (roughly the transport part). Their ratio tells you in advance whether this pair is reaction-dominated or transport-dominated at the chosen δ. Also compute the mass fraction within two voxels of the box walls after padding; it should be below 10^{-4}.

Part F. Porting to 3D and running today

The same three edits as in 1D, in `3d/setup/`; nothing in the DCT structure, the jitted loop or the mixed-precision path changes.

1. `state_3d.py`: add field `s` of shape `(nt, nx, ny, nz)` to the state (both copies), include it in the norm and the arithmetic.

2. `prox_3d.py`: `q2 = mx**2 + my**2 + mz**2 + s**2` (on the cell-centre copy), shrink `s` by the same factor as the momenta.

3. `projection_expsemi_3d.py` (and `projection_3d.py` if the CN path is kept): add `1/delta**2` to the diagonal of the per-mode time system in `precomp`, replace the DC-mode special case by the regular solve with `lambda_t + 1/delta**2`, subtract `s/delta` in the residual, add `phi/delta` to `s` in the output. The ETD coefficients are unchanged.

4. `pipeline_3d.py`: `A_fn`/`At_fn` identity on `s`; initial `s = 0`; return `ke`, `reaction`, `mass_t`.

5. **Tests before data**, in this order, at 32^3 then 48^3: T1 uniform $\to$ 2×uniform with $\varepsilon = 0.01$, $\delta = 0.3$ (cost $2\delta^2(\sqrt{2}-1)^2$, exact for any ε, converges in tens of iterations); T4 equal-mass Gaussians with $\delta = 0.5, 1, 2, 4$ against the balanced solver (difference $\propto 1/\delta^2$); T6 mass balance to machine precision. These three take minutes and catch every sign.

6. **Spheres.** Their synthetic case is public: five frames of a translating, diffusing Gaussian on 50^3 with mass factor $(1 + a_i\chi_i)$, $a = (0, 0.1, 0.2, 0.1, 0)$, χ_i a radius-1.5 ball around the centre. Run pair $0 \to 2$ with $\delta = \sqrt{9000}/50$, $\varepsilon = 3.2 \cdot 10^{-6}$ (their setting; the solver has run 10^{-3} at 96^3 and the 1D code handles $\varepsilon = 0$, so expect no difficulty) and compare the mass curve `mass_t` with $(1 + a_i)$ and the location of $\zeta > 0$ with the ball. Note their blur per frame, std $(i+1)\sqrt{0.2}$ voxels, is far more than $2\sigma T$ accounts for; the unbalanced solver will spend some of its source on that mismatch.

7. **Rat pair.** One early and one late pair as in Part E, at the padded 56^3 first. Report M_1/M_0, kinetic and reaction energies, the mass curve, a time-averaged g map alongside their Fig. (early $r > 0$ in CSF, later $r < 0$ at the skull base), and a speed map. Then 106^3 in `mixedc` precision if time allows.

8. **Write-up.** State the two parameter translations with their numbers, the hard versus soft endpoint difference, the padding, and that ε at their value is a regularizer while $\varepsilon_{\text{phys}}$ is 10^2 larger; give the mass curves and energy split for both ε values.

A Answers and hints

Ex. 1. v has units length/time, g has units 1/time, so δ is a length. The metric $ds^2 = |v|^2 + \delta^2 g^2$ needs a length to compare moving mass with changing it.

Ex. 2. With $\rho = \sigma^2$, $(\partial_t\rho)^2/\rho = 4(\partial_t\sigma)^2$, and $\int_0^1 (\partial_t\sigma)^2 dt$ with fixed endpoints is minimized by the straight line. Uniform ρ_0 stays uniform, so $\Delta\rho = 0$ and the diffusion term never acts.

Ex. 3. Naive crossover $\frac{1}{2}d^2 = 4\delta^2$, $d = 2\sqrt{2}\,\delta$. The exact geodesic shrinks mass while moving it and recreates it on arrival, with cost $\propto \delta^2(1 - \cos(d/2\delta))$, finite only for $d < \pi\delta$; beyond that the geodesic is pure teleport. In T3′ the optimizer does exactly that: destroys part of the mass at the start, moves the rest, and regrows it at the target, so the total mass dips in the middle.

Ex. 4. Lagrangian $\iint \frac{1}{2}\rho(|v|^2 + \delta^2 g^2) - \phi(\partial_t\rho + \nabla\cdot(\rho v) - \varepsilon\Delta\rho - \rho g)$; vary v: $\rho v = \rho\nabla\phi$; vary g: $\delta^2\rho g = \rho\phi$. So $\phi = \delta^2\zeta/\rho$ and $v = \nabla\phi$. As $\delta \to \infty$ with ϕ fixed, $\zeta = \rho\phi/\delta^2 \to 0$: the source is switched off and the constraint becomes the balanced one, with error $O(1/\delta^2)$.

Ex. 5. (a) $m/\rho + (m - \bar{m})/\sigma = 0 \Rightarrow m = \bar{m}\rho/(\rho + \sigma)$, same for s. (b) Substituting, the m, s part of the objective becomes $q^2/(2(\rho + \sigma))$; differentiating $q^2/2(\rho + \sigma) + (\rho - \bar{\rho})^2/2\sigma$ in ρ and clearing denominators gives the cubic. (c) It is the perspective of $\frac{1}{2}|(m, s)|^2$, jointly convex in (ρ, m, s) because the perspective of a convex function is convex. (d) $\rho\, h(\zeta/\rho)$ with $h(g) = g\log g - g + 1$ or $|g|$: the s-stationarity is no longer linear, so a scalar Newton iteration per cell replaces Cardano.

Ex. 6. (a) $MM^\top = A_{\mathrm{FP}}A_{\mathrm{FP}}^\top + (1/\delta)^2 I$ because the s block of M is $-I/\delta$. (b) $x = \bar{x} - M^\top\varphi$: the s component of $M^\top\varphi$ is $-\varphi/\delta$, so $s = \bar{s} + \varphi/\delta$. (c) The DCT diagonalizes the spatial part; $\delta^{-2}I$ is already diagonal in every basis and commutes with everything, so it lands on the diagonal of each mode's time system. (d) The Neumann time Laplacian has eigenvalues $\lambda_t \geq 0$ with one zero; $\lambda_t + \delta^{-2} > 0$. The DC component of the residual is the mass defect per time slice, and it is now absorbed by s instead of being required to vanish.

Ex. 7. A is block diagonal with an interpolation block of norm ≤ 1 and an identity block of norm 1. LADMM convergence needs $\tau \geq \gamma\|A\|^2$; with $\|A\| \leq 1$ the balanced choice $\tau = 1.01\gamma$ still works. s sits on cell centres in t and x because that is where the FP residual is evaluated, so the constraint couples them without interpolation.

Ex. 8. 3D: 5 fields instead of 4, so about +25% memory for the state and the same fraction in the prox and residual; the projection's DCT and time solves are unchanged in count. Iterations fall when the DC mode was the slow mode (it is now regular); they rise when δ is small enough that teleporting is optimal, since the problem then has flat directions in where exactly the mass dies and is reborn.

Ex. 11. ρ: tracer concentration proxy (percent signal change), masked. $\int\rho$: total tracer in the field of view. v: bulk cerebrospinal-fluid flow along perivascular spaces, the quantity their pathlines draw. ε: tracer diffusivity in tissue (Exercise 13 says theirs is far below physical). $g > 0$: tracer arriving, from the infusion or from outside the field of view, or apparent gain from signal saturation relaxing. $g < 0$: clearance to lymphatics and blood, tracer leaving the field of view. δ: the distance beyond which it is cheaper to clear and re-inject than to move; $\sqrt{\alpha} = 100$ voxels for the rat. $t \in [0, 1]$: the 10 minutes between the two frames.

Ex. 12. $\frac{1}{2}(\rho|v|^2 + \alpha\rho r^2) = \frac{1}{2}\rho(|v|^2 + \delta^2 g^2)$ with $\delta^2 = \alpha$. Under $x' = x/L$, $t' = t/T$: $v' = vT/L$, $g' = gT$, so $|v|^2 + \alpha g^2 = (L/T)^2(|v'|^2 + (\alpha/L^2)g'^2)$ and $\delta' = \sqrt{\alpha}/L$. Rat: $100/106 = 0.94$; spheres: $94.9/50 = 1.90$. Both are about one box length, so $\pi\delta$ exceeds the brain: the source will never teleport mass across the domain; it adjusts the total locally, preferring places where ρ is large because the cost is ζ^2/ρ.

Ex. 13. $\varepsilon = \sigma T/L^2$: $0.002 \cdot 4/106^2 = 7.1 \cdot 10^{-7}$ and $0.002 \cdot 4/50^2 = 3.2 \cdot 10^{-6}$. Units: $0.002\,\mathrm{voxel}^2/150\,\mathrm{s} \times 0.09\,\mathrm{mm}^2/\mathrm{voxel}^2 = 1.2 \cdot 10^{-6}\ \mathrm{mm}^2/\mathrm{s}$, about $100\times$ below the tissue diffusivity of Gd-DOTA. Physical: $\varepsilon = DT/L^2 = 1.5 \cdot 10^{-4} \cdot 600/(106 \cdot 0.3)^2 \approx 9 \cdot 10^{-5}$, so $\sim 10^{-4}$, inside the ranges already run. Run both: theirs to compare with their maps, the physical one because it is the value the paper's own "ε is tracer diffusivity" claim implies.

Ex. 14. (a) and (b) are genuine sources at the scale of the model (mass crossing the field-of-view boundary or the blood–CSF barrier); (c) and (d) are artefacts that the model will still book as ζ. The Fisher–Rao cost favours gain and loss where ρ is large, i.e. in the bright CSF spaces, consistent with their maps.

Ex. 15. (a) ζ absorbs it: pixel noise of amplitude η on a level ρ costs $\sim \delta^2\eta^2/\rho$ per cell per unit time, cheap where ρ is large and expensive in dim tissue, so noise in dim regions is instead transported in from neighbours. (b) Smoothing both frames with one kernel removes the

component that a soft endpoint would have ignored, without changing the solver. (c) Only the terminal row of the constraint.